

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/323772077>

Clustering of Patient Disease Data by Using K-Means Clustering

Article in *International Journal of Computer Science and Information Security*, · March 2018

CITATIONS

19

READS

4,741

1 author:



[Parasian Silitonga](#)

Universitas Katolik Santo Thomas Sumatera Utara

10 PUBLICATIONS 44 CITATIONS

SEE PROFILE

Clustering of Patient Disease Data by Using K-Means Clustering

Parasian D.P Silitonga

*Teknik Informatika, Universitas Katolik Santo Thomas Sumatera Utara
Jln. Setia Budi No. 479-F Medan, Sumatera Utara, Indonesia*

parasianirene@gmail.com

Abstract—Clustering is a method of grouping records in a database based on certain criteria. One method of clustering is K-Means Clustering. K-Means Clustering divides data into multiple data sets and can accept data inputs without class labels. This research uses K-Means Clustering method and implemented on patient disease data at Haji Adam Malik Hospital in Medan. The results of this study provide an illustration of the tendency of patient diseases at Haji Adam Malik Hospital. Through this research is expected to be a reference to anticipate priority services for patients, especially patients Social Security and Healthcare Security user.

Keywords—Clustering, K-Means Clustering, Social Security and Healthcare Security

I. INTRODUCTION

Clustering is a technique of grouping records on a database based on certain criteria. The clustering result is given to the end user to give an idea of what happens to the database [4]. Clustering performs grouping of data without any particular data class. Even clustering can be used to label the unknown data class. Therefore, clustering is often classified as an unsupervised learning method [3].

One method of data grouping is K-Means Clustering. The K-Means Clustering method divides the data into groups and can accept inputs of data without class labels [2]. This method partitions the data into clusters/groups so that data that have the same characteristics are grouped into the same cluster and data that have different characteristics are grouped into other groups.

Haji Adam Malik Hospital is one of the hospitals providing health care facility for the Social Security and Healthcare Security user. With the enactment of the Social Security and Healthcare Security program resulted in an increase in the number of patients. With the increasing number of hospital patients, it is expected that the hospital can know the classification of diseases that exist in the community and the cause of the disease. This can contribute input to the government of North Sumatra province in relation to disease prevention and health education to the regions.

The results of this study are expected to provide input for the government of North Sumatra province relating to the prevention of disease and health education to the regions, while for the hospitals provide input process anticipation priority services.

II. SOCIAL SECURITY AND HEALTHCARE SECURITY

Social Security and Healthcare Security is an institution established to organize social security program in Indonesia according to Law Number 40 of 2004 and Law Number 24 of 2011. According to Law Number 40 of 2004 regarding National Social Security System, Social Security and Healthcare Security is a non-profit legal entity [10].

The Social Security and Healthcare Security is a government-owned insurance agency established under Presidential Regulation No. 12 of 2013. The Social Security and Healthcare Security covers all Indonesian citizens and officially operates from 1 January 2014.

The purpose of implementing this program is to meet the appropriate health needs provided to every member of the community who has paid contributions or paid by the government or private institutions. With the existence of the Social Security and Healthcare Security program, all layers of society receive the right to health services from hospitals and health centers that have been appointed by the government as a hospital provider of Social Security and Healthcare Security.

III. DATA CLUSTERING

Data mining is a process used to find patterns of data tendency through a collection of data stored in storage by using pattern recognition techniques such as statistical and mathematical techniques [5].

One of the functions of data mining is clustering. Clustering is a technique of grouping records on a database based on certain criteria. The clustering result is given to the end user to give an idea of what happens to the database [4].

Clustering performs grouping of data without any particular data class. The clustering process can be used to label unknown data classes. Clustering methods include descriptive methods, and unsupervised learning where there is no class definition of the previous object [3]. Clustering groups a number of objects into clusters where good clusters have a high degree of similarity between objects within a cluster and a high degree of inequality with other cluster objects.

The clustering analysis identifies a collection of objects that resemble one another. A good clustering method can produce a qualified cluster to ensure the similarity of existing data in a cluster.

IV. K-MEANS CLUSTERING

K-Means is a fairly simple corticalization algorithm that partitions the dataset into several clusters k. The algorithm is fairly easy to implement and run, relatively fast, easy to customize and widely used.

The K-Means Clustering method divides the data into groups and can accept inputs of data without class labels [2]. This method partitions the data into clusters / groups so that data that have the same characteristics are grouped into the same cluster and data that have different characteristics are grouped into other groups.

This method is a non-hierarchical clustering method that classifies data in the form of one or more clusters. The data that have the same characteristics are grouped in one cluster and the data having different characteristics are grouped with another cluster so that the data in one cluster has small variation level [1].

K-Means Clustering method is performed as follows [2] :

1. Select the number of k clusters.
2. Initialization of k cluster center. In general, cluster centers are given initial values with random numbers.
3. Allocate all data / objects to the nearest cluster. The proximity of two objects is determined by the distance of the two objects. Likewise, the proximity of a data to a particular cluster is determined the distance between the data with the cluster center. To menghiutng distance all data to each cluster central point can use Euclidean distance theory.
4. Recalculate the cluster center with the current cluster membership. The cluster center is the average of all data / objects in a particular cluster.
5. Check each new cluster center user object. If the center of the cluster does not change anymore then the clustering process is complete. Or, go back to step 3 until the center of the cluster does not change anymore.

V. DISCUSSIONS

The object of the study was conducted on patient data sourced from medical record data of patient in RSUP Haji Adam Malik user of Security and Healthcare Security of 2014. The data obtained consist of Medical Record Number, Patient Name, Date of Entry, Date of Exit, Date of Birth, Age, Code of INACBG (Patient Disease Code) and Description of Patient Disease.

TABLE I
INACBG

Data Attribute	Information
InaCBG'sCode	InaCBG Data-Based Disease Code
Description	Disease Description

TABLE III
PATIENT_DATA

Data Attribute	Information
MedicalRecordNumber	Patient's Medical Record Number
PatientName	Patient's Name
DateOfBirth	Patient's Date of Birth
DateOfEntry	Patient's Date of Entry
DateOfExit	Patient's Date of Exit
InaCBG'sCode	InaCBG Data-Based Disease Code

A. Data Transformation

Stages of data transformation is the process of converting data into the appropriate form. The initial change is done by changing the data format so that the number of diagnoses of the patient's disease every month (January to December) is known. The result of data transformation performed in accordance with the table form in Table I.

TABLE IIIII
TRANSFORMATION DATA

Data Attribute	Information
InaCBG'sCode	InaCBG Data-Based Disease Code
Month	Calendar Month
Count	Number of Patient

The process of data transformation is done to obtain the trend of patient disease data every month during 2014, as well as to eliminate empty data. Results of data transformation process as in Table IV.

TABLE IVV
TABEL TRSANSFORMASI DATA PASIEN

No.	Code	January	February	...	December
1	A-4-10-I	0	0		2
2	A-4-10-II	9	12		15
3	A-4-10-III	13	14		13
...	...	...	...	...	...
574	Z-4-12-I	3	9		7

B. Clustering using WEKA

K-Means clustering process is done using WEKA software. WEKA consists of several tools that can be used to perform pre-processing tasks of data, classification, regression, clustering, association rules, and visualization [7].

Initialisation of the number of clusters is as much as 4 clusters with a maximum number of iterations of 1000 iterations. The result of the clustering process is obtained as in Fig 1 and Fig 2.

REFERENCES

- [1] Agusta. Y, "K-means - Application, Problems and Related Methods", Journal of Systems and Informatics, vol. 3, pp. 47-60, Feb. 2007.
- [2] Berkhin Pavel, *Survey of Clustering Data Mining Techniques*, Accrue Software, Inc : 2002.
- [3] Ian. H and Eibe Frank, *Data Mining Practical Machine Learning Tools and Techniques*, Morgan Kaufmann Publishers. San Francisco: 2005.
- [4] Jiawei Han and Micheline Kember, *Data Mining: Concepts and Techniques Second Edition* , Morgan Kaufmann Publishers. San Francisco: 2006.
- [5] Larose, Daniel, *Discovery Knowledge in Data*, A Jhon Wiley & Sons, Inc Publication. Canada: 2005
- [6] Manning, Christopher D., Prabhakar Raghavan, Hinrich Schutze, *An Introduction to Information Retrieval*, Cambridge University Press. Cambridge: 2009.
- [7] N. Sharma, A. Bajpai, and R. Litoriya, Comparison the Various Clustering Algorithms of WEKA Tools, *Int. J. Emerg. Technol. Adv. Eng.*, vol. 2, May 2012.
- [8] Presidential Regulation No. 12 of 2013 on Health Insurance.
- [9] Regulation of the President of the Republic of Indonesia No. 32 of 2014 on the Management and Utilization of National Health Insurance Capitation Funds at First Level Regional Government Owned Health Facilities.
- [10] Law Number 24 of 2011 on Social Security and Healthcare Security.

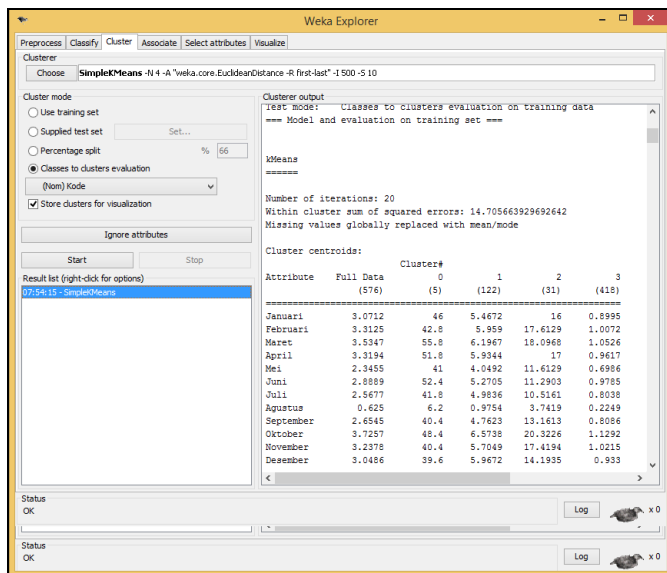


Fig. 1. Results of K-Means clustering process

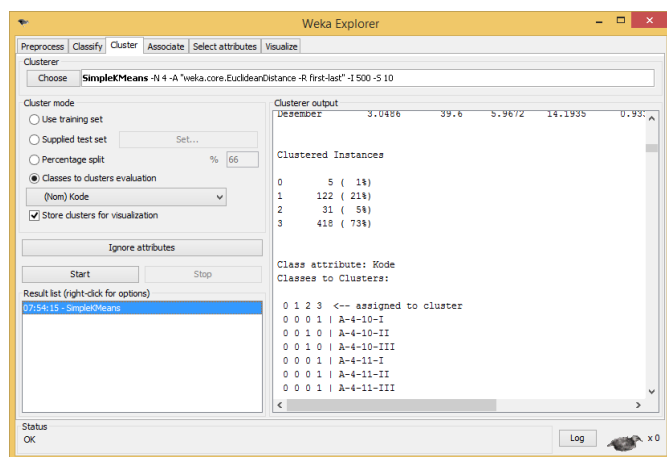


Fig. 2. Percentage of K-Means clustering results

VII. CONCLUSION

This research was conducted to find out the pattern of disease tendency of patients who went to Adam Haji General Hospital. Based on the research, it was found that Mild Septicemia Disease (A-4-10-I) is a disease with the highest tendency pattern. With the results obtained, it is expected that in the future the hospital can anticipate the priority of service based on the pattern of trends obtained.

ACKNOWLEDGMENT

The research was funded by the Ministry of Research, Technology, and Higher Education of the Republic of Indonesia. The authors would like to thank the Minister of Education of the Republic of Indonesia, Chancellor of St. Thomas Catholic University and Director of Haji Adam Malik General Hospital for all their support and encouragement.